

HOMEWORK #2 (Part: A)

- Reading for continuity:-

- Continuous only if graph have no breaks, jumps, or holes in that interval.
- exponential function, polynomial, and sine and cosine are continuous on every interval.
- Rational functions are continuous when their denominator $\neq 0$.
- Function created by adding, multiplying, or composing continuous functions are also continuous.

- Intermediate value theorem:-

- a given interval $[a, b]$
where $f(a) \neq f(b)$
 $f(a) < k < f(b)$
- "if the function is continuous, there should be a value 'c' where the function 'f' gives the answer 'k'."

for example:-

$$f(a) \neq f(b) \quad [0, 5]$$

$$f(x) = x^2 + x - 1$$

$$f(a) = 0^2 + 0 - 1 = -1$$

$$f(b) = 25 + 5 - 1 = 29$$

for $k = 11$ find "c":-

$$-1 < k < 29$$

There should be a value of $f(x)$ where $f(x) = k$:-

$$f(c) = k$$

$$11 = x^2 + x - 1$$

$$0 = x^2 + x - 12$$

$$0 = x^2 + 4x - 3x - 12$$

$$0 = x(x+4) - 3(x+4)$$

$$0 = (x+4)(x-3)$$

$$\boxed{\begin{matrix} x = -4 \\ x = 3 \end{matrix}}$$

$$c = -4, 3$$

Home work #2 part (B) reading

Speed is the magnitude of velocity

average velocity = $\frac{\text{change in position}}{\text{change in time}}$ or $\frac{f(a+h)-f(a)}{a-h}$

Instantaneous velocity = $\frac{s(a+h)-s(a)}{h}$ where limit $h \rightarrow 0$

Home work 2:- PART (B)

EXERCISE 3 problems for section 2.1:-

Data:-

Q1:- distance = 5

time = t

avg velocity = ?

Solution:-

$$\text{avg velocity} = \frac{f(5) - f(2)}{5 - 2}$$

$$= \frac{400 - 135}{3} = \frac{265}{3} = 88.33 \text{ m/s}$$

Result:-

The average speed/velocity of car during $t=2$ to $t=5$ is 88.33 m/s .Q4:- Data:-at $t=0$ the distance traveled = 1m.at $t=3$ the distance traveled = 4m

Solution:-

$$\text{avg velocity} = \frac{f(3) - f(0)}{3 - 0} = \frac{4 - 1}{3 - 0} = \frac{3}{3} = 1 \text{ m/s}$$

Result:-

The particles average value is 1 m/s from $t=0$ to $t=3 \text{ sec}$.

Q6:- Data:-

$$\text{distance in micrometers} = s(t) = e^t - 1$$

average velocity of particle from $t=2$ to $t=4$

Solution:-

$$\text{for } s(2) = e^2 - 1 = 6.389 \mu\text{m}$$

$$\text{for } s(4) = e^4 - 1 = 53.598 \mu\text{m}$$

converting it in meter:- $6.389 \times 10^{-6} \text{ m}$ and $53.598 \times 10^{-6} \text{ m}$

Now:-

$$\text{average velocity} = \frac{s(4) - s(2)}{4 - 2} = \frac{53.598 \times 10^{-6} - 6.389 \times 10^{-6}}{2}$$

$$\text{avg velocity} = 2.36 \times 10^{-5} \text{ or } \boxed{23.6 \mu\text{m/s}}$$

Result:-

The average velocity of particle from $t=2$ to $t=4$ is $23.6 \mu\text{m/s}$.

Q8:-

Data:-

$$\text{distance} = S = 3t^2$$

$$\text{distance at } t=1:- S = 3(1)^2 = 3 \text{ m}$$

$$\text{distance at } t=1+h:-$$

$$\text{where } h=0.1 \text{ so, } t=1.1:-$$

$$h=0.1: S = 3(1.1)^2 = 3.63 \text{ m}$$

$$h=0.01: S = 3(1.01)^2 = 3.06 \text{ m}$$

$$h=0.001: S = 3(1.001)^2 = 3.003 \text{ m}$$

a) average velocity=?

b) - Instantaneous velocity at $t=1$.

a)

i) $t=1$ to $t=1.1$

$$\text{average velocity} = \frac{s(1.1) - s(1)}{1.1 - 1} = \frac{3.63 - 3}{0.1} = 6.3 \text{ m/s}$$

ii) $t=1$ to $t=1.01$:-

$$\text{average velocity} = \frac{s(1.01) - s(1)}{1.01 - 1} = 6 \text{ m/s}$$

iii) $t=1$ to $t=1.001$

$$\text{average velocity} = \frac{s(1.001) - s(1)}{1.001 - 1} = 2 \text{ m/s}$$

b)

instantaneous velocity at $t=1$

$$v = \frac{s(1+h) - s(1)}{h}$$

$$v = \frac{3(1+h)^2 - 3(1)^2}{h}$$

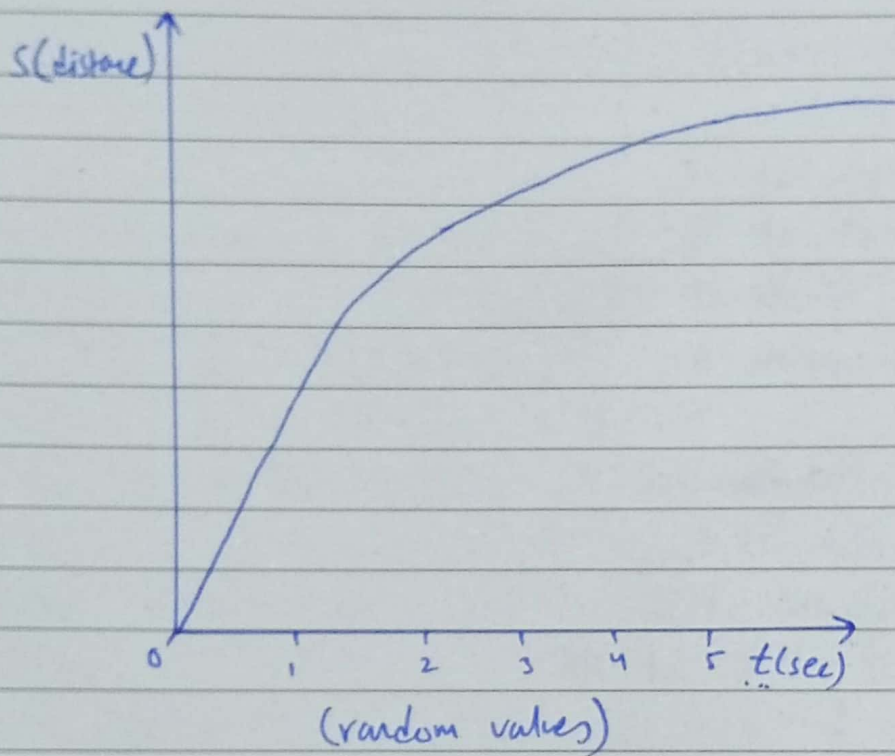
$$v = \frac{3(1^2 + 2h + h^2) - 3}{h} = \frac{\cancel{3} + 6h + 3h^2 - \cancel{3}}{h} = \boxed{3h + 6 \text{ m/s}}$$

$\lim_{h \rightarrow 0} = 3(0) + 6 = 6 \text{ m/s}$

Result:-

- a) The average velocities at respective times are 6.3 m/s, 6 m/s, 2 m/s
 b) The instantaneous velocity at $t=1$ is equal to 6 m/s

Q15:- starts at high speed.
then decreases slowly.



Q18:- Match the slopes:-

slope	point
-3	F
-1	C
0	E
$1/2$	B
1	A
2	D

Q20:-

Given-

$$y=f(x)$$

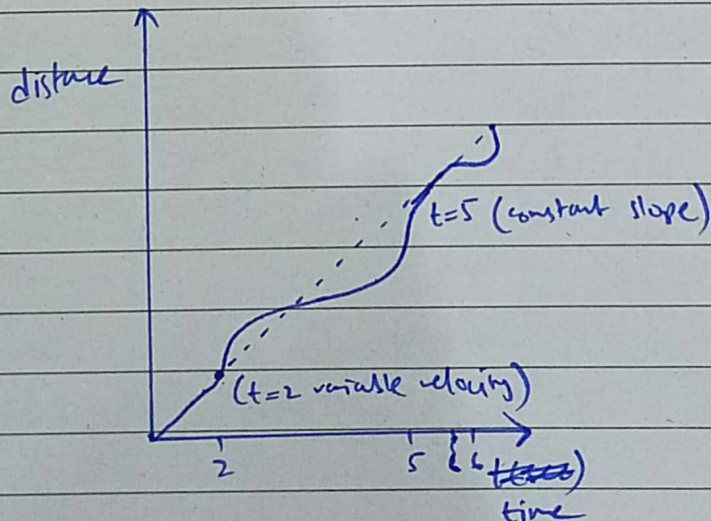
solution:-

- The number zero.
- The number 1
- slope of graph at C
- Slope of graph at B
- The slope of ~~graph~~ ^{line} at AB $\leftarrow 1$
- The slope of graph A

- The number zero
- slope of graph at C
- slope of graph at B
- The slope of line at AB
- The number 1
- The slope of graph A.

Q23:-

average velocity from $t=2$ to $t=6$ is equal to instantaneous velocity at $t=5$



Q24.

a)

$$y = 0$$

$$0 = -16t^2 + 50t + 36$$

using quadratic equation.

$$t = 3.72 \text{ sec.}$$

Now, $S = vit + \frac{1}{2}at^2$

$$\therefore v_i = 0$$

$$S = \frac{1}{2}(9.8)(3.72)^2$$

$$S = 4.9 \times 13.83 = 67.80 \text{ meters feet.}$$

Ans:- a):-

at time $t=0$ the ball is at the height of bridge.

so

$$f(0) = 0 + 0 + 36 = \boxed{36 \text{ feet.}}$$

b) average velocity of ball in first second =

$$f(0) = 0 + 0 + 36 = 36 \text{ feet.}$$

$$f(1) = -16(1)^2 + 50(1) + 36 = 70 \text{ feet.}$$

$$\text{average velocity} = \frac{f(1) - f(0)}{1 - 0} = \frac{70 - 36}{1} = \boxed{34 \text{ feet/sec.}}$$

c) at $t=1$

$$\text{instantaneous} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \frac{-16(1+h)^2 + 50(1+h) + 36 - 70}{h}$$

$$= \frac{-16(1+2h+h^2) + 50 + 50h + 36 - 70}{h}$$

$$= \frac{-16 - 32h - 16h^2 + 50 + 50h + 36 - 70}{h}$$

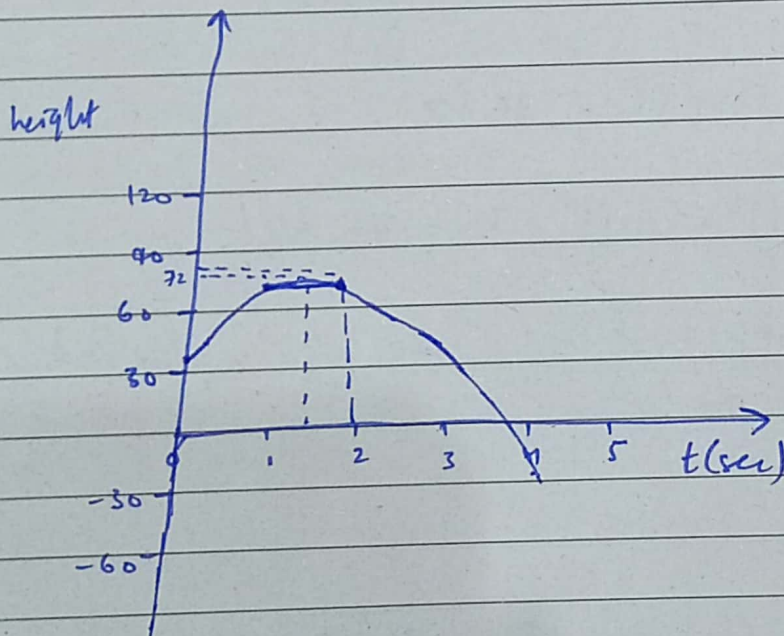
$$\frac{-16h^2 + 18h + 70}{h}$$

$$\text{Instantaneous velocity} = \frac{-16h^2 + 18h}{h} = -16(h) + 18$$

$$\text{Now } h \rightarrow 0 = -16(0) + 18 = \boxed{18 \text{ feet.}}$$

d) $t = 0 \quad 1 \quad 2 \quad 3 \quad 4$
 $f(t) = 36 \quad 72 \quad 72 \quad 42 \quad -20$

d) $t = 0 \quad 1 \quad 2 \quad 3 \quad 4$
 $f(t) = 36 \quad 70 \quad 72 \quad 42 \quad -20$



The max height which ball reached is ~~72~~ ^{vertex of parabola} 75.06 m and the velocity at that time was zero.

$$\text{max height} = \frac{-b}{2a} = \frac{-50}{2(-16)} = 1.56 \text{ sec.}$$

$$y = f(1.56) = -16(1.56)^2 + 50(1.56) + 36 = \boxed{75.06 \text{ m}}$$

c): At time $t = 1.56 \text{ sec}$ the ball reaches its maximum height of 75.0 m as shown in the graph. After that the ball again started to move downwards.